Depth, Water, and Oak Gall

Living ink, next law. Four changes: paper depth and dents, ink in depth with a side view, the water brush, and oak gall ink. Build spec for the coding agent.

The principle all four follow

> **Nothing computes unless it is touched or viewed. A tile runs only while it has wet ink. Time is applied to a record only when someone looks at it, in one step, never tick by tick while nobody is there.**

This is the same rule as the sleep fix for the long-stroke lag: dry tiles leave the active set and only a new deposit wakes them. The new changes add more kinds of matter and more ways for tiles to wake, but nothing that runs forever. Oak gall darkening is the case where this matters most, and section 4 shows how it stays at zero cost while unviewed.

Law version

Law v1 is frozen. All four changes go into `law/v2/`, a new frozen package with its own goldens. Records carry their law number: v1 records keep rendering under v1 exactly as before, and new writing is made under v2. Nothing about a v1 sheet changes when v2 ships.

1. Paper depth and dents

The paper gets a short Z axis. It stays a 2D grid of cells, but each cell now has a surface height that writing can push down.

- **Dent.** Each cell gets a new field `dent` (Uint8, starts at 0). On deposit, the nib footprint pushes the surface down by an amount from pressure × dwell: `dent += DENT_Q[p] × dt`, clamped to `DENT_MAX`. Plastic: a dent never recovers. With no real pressure, synthetic flex p and dwell drive it, as for width.
- **A dent is a groove.** It feeds into flow two ways: it lowers the cell's height in ϕ (so surface ink runs into it), and it raises the cell's capacity (`cap += dent × DENT_CAP`, so the groove holds more ink). Real paper does this: scratched lines pool ink.
- **Fibers compress.** In a dented cell, sizing (how fast it drinks) drops a little, so ink in a groove sits wet longer and settles darker.
- **It is a fact.** The dent map is part of the settled state and goes into `factsDigest`. It comes only from the acts and the paper, so it regrows byte for byte like everything else.

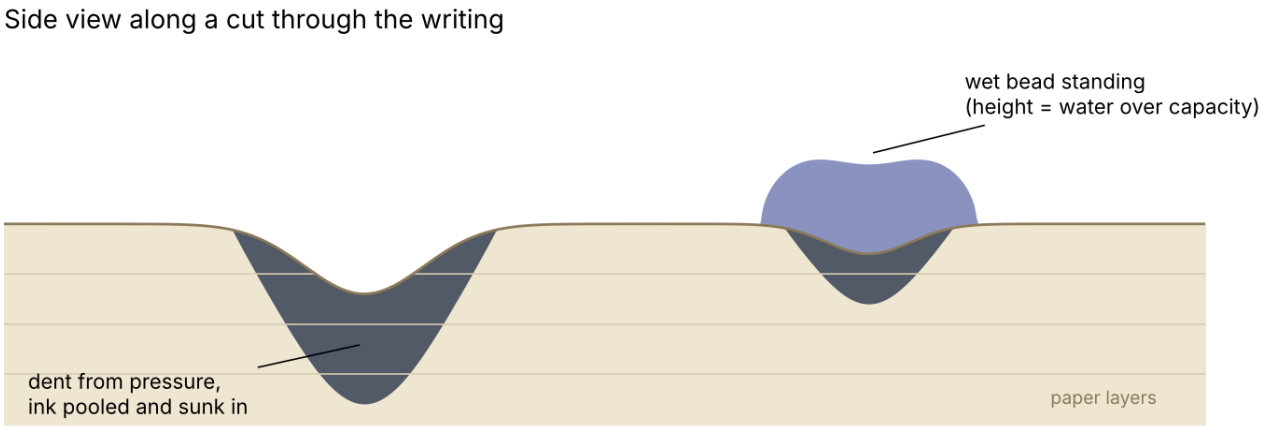
2. Ink in depth: beads, sinking, and the side view

The sim already has height: wet water over capacity is the bead. Two things are added so depth can be seen.

- **Penetration.** Each inked cell gets `depth` (Uint8): how far settled ink sank into the fibers. During settle, `depth += SETTLE_DEPTH[siz] × settled_amount`, clamped to the paper thickness. Thirsty cells and grooves sink deeper.
- **Peak height.** Each inked cell records `peak` (Uint8): the highest the wet bead stood over it before drying. That keeps the droplet visible in the side view after it has dried.
- Both are facts, in the digest.

The side view

A viewer mode, not part of the published look. Pick a line across the writing; the view shows that cut: the paper body in layers, the surface with its dents, ink sunk into the fibers below the surface, and the bead above it (live while wet; from `peak` after drying).



The relief view

The top view lit from one side, using dent and bead height as a height map, so grooves and dried droplets read as depth on the page. Also a viewer mode. The published look stays flat, as ruled before (no height shading in published looks).

- Both views read the same facts, so they work for any sheet the viewer can see: own and open sheets regrow on the device, and closed sheets are served by the render pool like looks (`/look?mode=side&cut=...` and `mode=relief`).

3. The water brush

A pen kind that carries water and no pigment. Use it over dried ink and the ink comes back to life and bleeds again.

- **Pen.** `kind = water` in the pen act. Deposit adds W only; pigment supply is 0. Same nib shapes, same flow and dwell behavior.
- **The react: water + fixed ink → wet ink.** In any cell with wet water and fixed pigment, each dry tick returns a fraction of the fixed ink to wet: ΔP `= F × REDISSOLVE[ink] × W / W_SAT`. Then normal flow carries it, so the old line feathers and blooms where the water spread.
- **Solubility is an ink property.** `REDISSOLVE` is per ink: ordinary inks revive easily; oak gall barely revives once it has darkened (true to life).
- **Waking.** A water deposit wakes the tile like any deposit. Dormant tiles that hold fixed ink come back into the active set, run until dry, and sleep again.
- **Scope in this build:** the water brush works on writing in the current draft. Reviving ink on an already sealed sheet needs a new record that starts from the old sheet's facts (see Needs ruling).

4. Oak gall ink

Iron gall ink goes on pale and darkens over days as the iron complex oxidizes in air. Two ways to write with it:

- **Gall ink (one part).** An ink whose pigment includes a gall component `G`. It writes pale gray-blue and settles as fixed `FG`.

- **Two-part (the react).** An iron pen and a gall-extract pen, each nearly clear. Where their wet inks meet in the same cell, they form gall: `ΔG = min(Fe, T) × G_REACT` per dry tick, consuming both. Write a line in one, cross it with the other, and black forms only at the crossing. This is the first when/then reaction in the law: when iron and tannin are wet in the same cell, then gall.

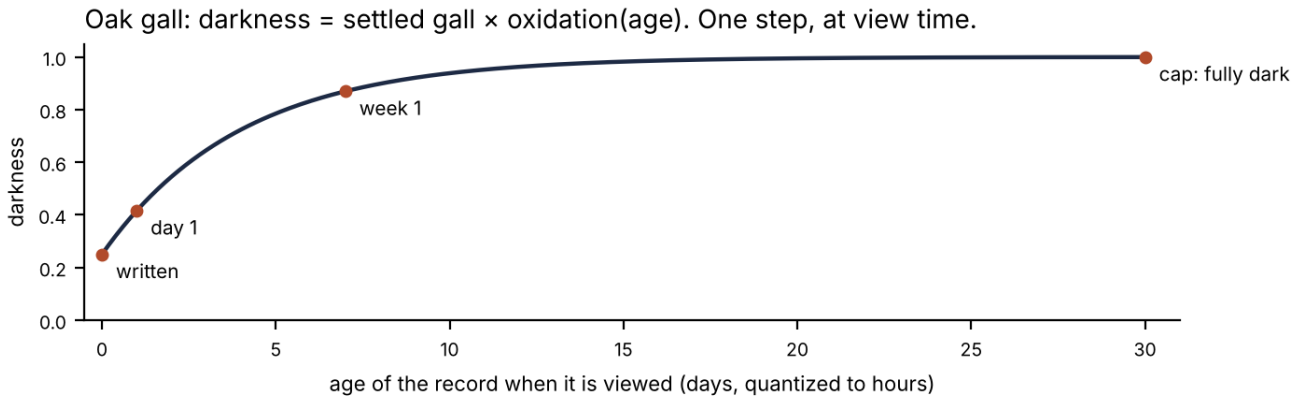
Darkening happens only when someone looks

Oxidation runs for days. Simulating it tick by tick would mean computing every gall sheet forever, viewed or not. Instead, darkening is a closed formula applied at view time:

```
settled facts (sealed, time-free):   FG per cell   (how much gall settled there)

shown darkness at view time:        dark = FG × OX[age]
                                    age  = view_time − record.made, quantized to hours
                                    OX   = literal table, 1 − e^(−age/τ), capped at AGE_CAP (30 days)
```

- **Zero cost while unviewed.** Nothing is stored or scheduled. A sheet nobody opens for a year costs nothing that year.
- **One step at view.** When it is opened, the age is computed and every cell is darkened in a single pass. Fast-forwarding a year costs the same as a day.
- **Still exact.** The facts (FG) are sealed and time-free, so verification never depends on when it is checked. The look at a given age is a pure function of facts + age bucket, so anyone gets the same image for the same age. Looks are cached per (ink, scale, age bucket).
- **Playback** can show the darkening as a time-lapse at the end, from pale to the age it has reached.
- **Inside a writing session** the oxidation curve is applied from the record's start the same way, so ink laid early in a long letter is a touch darker than ink laid at the end.



State and data cost

```
per cell, v1:      W, P0..2 (wet)   F0..2 (fixed)                    7 integers
per cell, v2 adds: dent, depth, peak (Uint8 each)                   +3 bytes
gall, sparse:      wet Fe, T, G and fixed FG, allocated only on tiles
                   that have ever held iron/gall ink                  +4 on those tiles
```

- Acts do not grow: the water brush and gall inks are pen acts (`kind`, ink id), already in PNKA's pen fields. A minute of writing is still about 20 KB.
- Nothing about depth or darkening is stored; it all regrows from the acts and the paper. The age is computed from the record's own `made` time.

Tests to add

- **Dent:** a hard slow stroke leaves a deeper dent than a light fast one; ink later laid across the groove pools in it (more fixed ink in the dented cells than beside them).
- **Depth:** thirsty cells sink deeper than sized cells; a dwell shows peak height in the side view after drying.
- **Water:** water over dried ordinary ink re-wets and bleeds (fixed decreases, wet appears, ink spreads past the old edge); water over darkened gall barely moves it; dormant tiles wake and go dormant again.
- **Two-part gall:** iron alone and tannin alone stay pale; black forms only where they crossed while both wet.
- **Lazy darkening:** the look at age 0, 1 day, 7 days, 30 days matches the table; the look at 400 days equals the 30-day cap; facts digest identical at every age; no timer, job, or write happens for an unviewed gall sheet.
- **Sleep rule:** a 60-second stroke with dents and depth still has flat per-tick cost.
- **Versions:** every v1 golden still renders identically under the v1 package.

Needs Tabor's ruling

- Water on sealed sheets: a new record that starts from an old sheet's facts (a continuation sheet), or water only within drafts.
- Darkening time τ and the cap (how many days to fully dark).
- Whether the side view and relief view are shown to visitors, or only to the writer and on open sheets.
- Paper thickness (how many depth levels), and dent strength for mouse and thumb with no real pressure.